Task 3 — Reflection on the AI Paradoxes

FTEC5660 Homework 2

This homework is deliberately two-faced: first you build an agent you can trust to verify people, then you turn around and break it. Doing both back-to-back makes several AI paradoxes concrete rather than abstract.

1. The autonomy–trust paradox

The very features that make an agent useful for KYC are the same features that make it attackable. A plain script that compares two fixed JSON records with hard-coded rules is narrow and brittle, but it is also impossible to *persuade*: you cannot flatter it, invoke authority on it, or hand it a fake "system" message. A tool-calling LLM agent, in contrast, can resolve duplicate names, understand that "BSc" and "Bachelor of Science" are the same degree, and decide which fields even matter — but that openness to language is precisely an input surface an adversary can write into. **Power and trust scale together.** The more autonomously an agent reads and reasons about untrusted documents, the more capable it is — and the more a carefully crafted document can steer it. Giving an AI more agency does not just add ability; it adds a target.

2. The accuracy ≠ robustness paradox

On the seven honest public CVs my verifier scored 7/7. A clean, high accuracy like that usually reads as "the system works." Yet the adversarial CV — which a human flags in about two seconds — also scored 0.95. High accuracy on normal inputs coexisted with near-total failure on a deliberately shaped input, and the "attack" did not even require outsmarting the model: it only required hiding one word from the text extractor while leaving it visible to the eye. Performance on the average case told me almost nothing about performance on the adversarial case. This is the central lesson of adversarial machine learning: **a model can be both impressively right and dangerously wrong at the same time**, and the failure is invisible to anyone who only measures ordinary accuracy.

3. The channel / grounding paradox

An agent is told to "trust the tools, not the document." But the boundary between *evidence* and *untrusted data* is enforced only in software, not in reality. The PDF, the extracted text, and the tool results all eventually arrive as tokens in the same model context. My first attack tried to impersonate that boundary with words ("SYSTEM", "pre-verified", a forged assistant turn); my defender, which had been told the CV is untrusted, correctly ignored it. My second attack skipped persuasion entirely and exploited the *physical* boundary between pixels and text — a gap the model cannot

reason about because it never sees the rendered page at all. The paradox: **the agent is asked to judge a visual artefact it cannot see**, through a lossy pipeline (PDF → text) whose blind spots are invisible to the model. Grounding an LLM in "real" tool data helps, but every layer between the world and the model — OCR, PDF parsing, tokenisation, prompt wrapping — is a place where "what is true" and "what the model receives" can quietly diverge.

4. The defender–attacker co-evolution paradox (and why red-teaming is free)

Building the verifier taught me exactly how to defeat one, and building the attack taught me exactly what the verifier failed to check. The two tasks are not opposites so much as two views of the same knowledge: every defence I wrote ("treat the CV as untrusted, strip delimiters, compare years exactly") named an attack, and every attack I attempted named a missing defence ("the loader does not OCR images, so compare against the rendered page"). Security here is not a feature you finish; it is an arms race in which the attacker only has to find one gap while the defender must close every one — an asymmetric, moving target. The encouraging part is that the same person, with the same model and an afternoon, can run both sides. Red-teaming is not extra work after the system is built; it is the cheapest way to discover what "trustworthy" actually means.

5. The human-oversight paradox

Automated KYC is deployed to scale human judgement and to reduce fatigue and bias. But the more reliable the automation appears, the more a human reviewer defers to it (automation bias) — and the adversarial CV is engineered for exactly that moment of deference. So automation intended to strengthen oversight can, paradoxically, hollow it out: the human remains nominally "in the loop" while no longer really looking. The honest conclusion is not "do not automate," but that oversight has to be designed for the cases the machine structurally cannot see — for instance, having a reviewer inspect the *rendered* document and its raw extraction side by side, and treating a perfect automated verdict as the start of scrutiny rather than its end.

Bottom line

The through-line is a single paradox: **the qualities we want from an AI agent — autonomy, language understanding, broad tool access, confident end-to-end judgement — are inseparable from the qualities that make it manipulable.** Task 1 measures how helpful that bargain can be; Task 2 measures its price; and the job of an engineer is to decide, per field and per risk, where that trade is actually worth making.